軟體測試

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授課時間：每週二 9：10—12：00 識別碼：921 U9550

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期中考

姓名：

科系年級：

學號：

考試中，請勿參閱資料。

1. Please explain Beizer testing level 2 and level 3. Please also give examples in practices for their differences. (10 points/10)

This is a closed-book exam.

2. Explain why edge coverage subsumes node coverage. Please give an example to show why a test set that satisfies edge coverage may not reveal the bugs that a test set satisfying node coverage may reveal. (10points/20)

3. Answer questions (a)–(d) for the graph defined by the following sets:
   - N = {0, 1, 2, 3}
   - N0 = {0}
   - Nf = {2}
   - E = {(0, 1), (0, 2), (1, 0), (1, 3),(3,2) (2, 0)}

   Also consider the following (candidate) paths:
   - p0 = [0, 1, 2, 0]
   - p1 = [0, 2, 0, 1, 3, 2]
   - p2 = [0, 1, 3, 2, 0, 1, 0, 2]
   - p3 = [1, 2, 0, 2]
   - p4 = [0, 1, 2, 1, 2]

(a) Which of the listed paths are test paths? Explain the problem with any path that is not a test path. (5points/25)

(b) List the test requirements for edge-pair coverage (only the length two subpaths). (5points/30)

(c) Does the set of test paths (part a) above satisfy edge-pair coverage? If not, identify what is missing. (5points/35)

(d) Consider the prime path [2, 0, 2] and path p2. Does p2 tour the prime path directly? With a sidetrip? (10points/45)

4. Below is a graph, defined by the sets of nodes, initial nodes, final nodes, edges, and defs and uses. Each graph also contains a collection of test paths. Answer the following questions about each graph.

Graph: N={0,1,2,3,4,5,6,7}

N0={0}

Nf={7}

E={(0,1),(1,2),(1,7),(2,3),(2,4),(3,4),(4,5),(5,6),(6,4),(6,1)}

def(0)=def(3)=use(5)=use(7)={x}

Test paths:

t1=[0,1,7]

t2=[0,1,2,4,5,6,1,7]

t3=[0,1,2,3,4,5,6,1,7]

t3=[0,1,2,4,5,6,4,5,6,1,7]

t4=[0,1,2,3,4,5,6,4,5,6,1,7]

t5=[0,1,2,3,4,5,6,1,2,4,5,6,1,7]

(a) Draw the graph. (5pts/50)

(b) List all of the du-paths with respect to x. (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/55)

(c) For each test path, determine which du-paths that test path tours. For this part of the exercise, you should consider both direct touring and sidetrips. Hint: A table is a convenient format for describing this relationship. (5pts/60)

(d) List a minimal test set that satisfies all-defs coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/65)

(e) List a minimal test set that satisfies all-uses coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/70)

(f) List a minimal test set that satisfies all-du-paths coverage with respect to x. (Direct tours only.) Use the given test paths. (5pts/75)

5. Construct two separate use cases and use case scenarios for interactions with a vending machine that sells coke, sprite, and orange juice. Do not try to capture all the functionality of the vending machine into one graph; think about two different people using the vending machine and what each one might do. Design test cases for your scenarios. (10pts/85)

6. For the following questions (a)–(c), consider the method FSM for a (simplified) programmable thermostat. Suppose the variables that define the state and the methods that transition between states are:

```
partOfDay : {Wake, Sleep}
temp : {Low, Med, High}
// Initially "Wake" at "Low" temperature
// Effects: Advance to next part of day
public void advance();
// Effects: Make current temp higher, if possible
public void up();
// Effects: Make current temp lower, if possible
public void down();
```

(a) How many states are there? (5pts/90)

(b) Draw and label the states (with variable values) and transitions (with method names). Notice that all of the methods are total. (5pts/95)

(c) A test case is simply a sequence of method calls. Provide a test set that satisfies edge coverage on your graph. (5pts/100)